

Standalone Kinematic Evaluation of a 5-DoF Robotic Manipulator via MATLAB Simulation

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Abstract—A strong grasp over simulation software is essential to develop practical working robots for optimal functioning. This paper aims to discuss the simulation-based design of 5-DoF Robotic manipulator focusing on the kinematics aspects i.e. forward, and inverse kinematics of the manipulator. Here the forward kinematics senses the position as well as the orientation of the end effector while inverse kinematics calculates the joint angles for the particular position of the end effector. Mechanical parameters, such as joint position, velocity, torque, acceleration etc. can be computed via this experiment. However, this study focuses exclusively on position because the other quantities are derived quantities of the fundamental quantity, “Position”. The objective of this paper is to provide an approach to feed a robotic manipulator with inputs as position vectors, then obtain the end effector’s position as the output and via forward and inverse kinematics and evaluate the accuracy in traversing the given set of inputs (position vectors). Thus, this study provides an overview of the necessities of the standalone functioning of the forward and inverse kinematics and the precision with which the plant operates with the kinematics as a standalone operation without the intervention of any control systems.

Index Terms—Forward Kinematics, Inverse Kinematics, Robotic Manipulator, 5-DoF Robot Arm, Simulation, End Effector Position, Position Vector, Kinematic Modeling, Robot Simulation Software, Standalone Operation

I. INTRODUCTION

In view of current technological advancements, the need for robotic manipulators with accurate and precise movement have increased and so is the competition among the various firms that produce such robots. Hence, there comes the need for simulation analysis through which feasibility, precision, accuracy, and performance can be effectively tested through repeated experiments until optimal conditions are met. With these conditions, simulated robotic designs can be translated into practical robotic systems. Through this study we demon-

strate methods to calculate the kinematics and dynamics of robotic systems in simulation.

A. Kinematics :

Kinematics is the study of the motion of any system without considering the cause of the motion. Meaning, it is the study of the effect and not the cause. It involves the computation of various mechanical parameters such as position, velocity, and acceleration, in both linear and angular cases along with time without considering the force and torque. In the field of robotics, it is categorized into two, forward kinematics and inverse kinematics respectively. Forward kinematics in robotic systems involves determining the position and the orientation of the end effector in 3D space upon the given joint variables, which can be either angle for revolute joints or linear displacements for prismatic joints. The DH parameter is used to achieve this systematically assigning coordinate frames to each link and construct a series of homogeneous transformation matrices that describe the relative motion between adjacent links. Inverse kinematics is the reverse process of forward kinematics where the solution is obtained by first acquiring the end positions of the links and solving the corresponding joint parameters. Existence of solutions for the inverse kinematics indicates the reachable workspaces and hence the stability of the position of the robotic arm while the non-existence shows that the workspace is dexterous, conveying that the position fed to the plant is unstable.

II. LITERATURE REVIEW

Wei-chen Lee and Shih-an Kuo [1] aims to build a simulation model robot considering that it is highly challenging to develop real world system before any prior knowledge of the functioning of the system. The intention was to create a

simulation model, conduct all various experiments to analyze the working efficiency and the feasibility of the plant before the real-world deployment. MATLAB/Simulink was used to integrate kinematics and motion control to perform the simulation. The simulator programs were then fed to the real robot and controlled through TwinCAT software, and the simulation results were verified. Bin Guo [2] proposes an exhaustive simulation-based study on a multi-Degree of Freedom robotic arm. The work qualitatively explains the fundamental principles like robotic kinematics, Workspace concepts (dexterous and reachable), Trajectory planning methods, and calculation of mechanical parameters like end effector's position, torque, velocity, acceleration etc. The study enhances the understanding of the robotic arm structural parameters that provides scope for development and deployment of intelligent robotic systems. Juan Liu and Qiang Luo [3] discusses the investigation of modeling and simulating robotic arm for polishing purposes in industries. MATLAB and Robotics toolbox are the platforms where the kinematics and dynamics of the multiple DOF robot is established. In addition, the paper deals with standard D-H modeling methods. The forward dynamics is used and the joint torque, joint angle, joint velocity, joint angular position etc., were computed and recorded, indicating the higher reliability and performance during polishing works. Dinh Tho Long [4] et al. introduces the simulation of kinematics and dynamics model of multi DOF robotic arm using MATLAB tool and Robotics toolbox. The forward kinematics of multi-step free controller is analyzed and the joint angle of the robotic arm is calculated using inverse kinematics- deducing the dynamics solution of the controller. The effectiveness of the simulation is presented through the computation of such parameters. Le Tien Dung et al. [5] designed a dynamic Sim-Mechanics model of a two-link planar robotic manipulator and integrated with PD control. The practical situation of Gravity is imparted to the design with online gravity compensation. Their simulation results demonstrated that the model performs as good as with constant gravity in the real world, offering improved accuracy in trajectory tracking and control on the motion. Željapah [6] discussed the importance of simulation in robotics for designing, testing, and improving robot systems. The paper explains how tools like MATLAB/Simulink help in modeling robot kinematics and dynamics and compares different simulation methods. It also highlights how simulation is used in areas like humanoid robots, mobile robots, space robots, and medical robotics. González-Palacios et al. [7] proposed an interface for the simulation of robots called SnAM (Serial n-Axis Manipulators). It was constructed using the ADEFID (Advanced Engineering platform for Industrial Development) framework which enables easy solving of both the forward and inverse kinematics. It helps users with various specifications such as planning of trajectory and collision avoidance. When the robot prototypes are integrated with the SnAM, it shall help in testing the performance of various designs. Zamanzadeh et al. [8] successfully modelled the 4-DoF articulated robotic arm and also both the forward kinematic and inverse kinematic model. The study found the correlation between the forward

and inverse kinematic model, and the performance of the kinematic model was also evaluated on the simulated robotic arm. The theoretical framework for further research is satisfied by the developed kinematic model. The robot arm trajectory planning, and structural optimization of designed robot were also calculated using the models. Hai Linh et al. [9] proposed that the real-time execution of motion control in robotic arms necessitates efficient computation of both forward and inverse kinematics. Traditional software-based implementations of these algorithms are computationally intensive and can significantly degrade system performance, particularly in complex articulated robotic systems. To address this limitation, a hardware-based solution for implementing forward and inverse kinematics in a five-axis articulated robotic arm has been given in the paper using VHDL.

Rokbani et al. [10] proposed a altered particle swarm optimization (PSO) method to solve the inverse kinematics (IK) of a 5-DOF 3D-printed robotic arm. Unlike standard PSO, m-PSO adapts particle behavior based on fitness, improving convergence. A multi-objective version (MO-m-PSO) optimizes both end-effector position and orientation using Pareto front strategy, with forward kinematics modeled via the D-H method. Yao et al. [11] addressed inverse kinematics in seven-DOF redundant manipulators, a method using virtual arm angle constraints is proposed. The study introduces an arm angle optimization approach that considers joint motion range as constraints to determine a feasible arm angle interval. It then computes the arm angle ensuring optimal manipulability based on the target pose and calculates the joint angles analytically. This method enhances solution stability and Cartesian space operability. Zarrin et al. [12] present the design, fabrication, and kinematic analysis of a five-DOF robotic arm for pick-and-place tasks. SolidWorks and SimMechanics were used for modeling and motion analysis, while plexiglas were selected to reduce weight and power consumption. A novel geometric inverse kinematics method was introduced, incorporating both end-effector position and orientation as inputs, improving workspace coverage and reducing computational error. The approach calculates joint angles from global end-effector coordinates using robot geometry. Experimental validation showed strong agreement between simulation and physical results, confirming reduced positioning errors and enhanced system efficiency. Chen et al. [13] addresses the limitations in precision caused by structural errors in industrial robotic arms, particularly those using spherical wrists. It introduces a kinematic modeling and configuration design framework aimed at enhancing inverse kinematics accuracy under real-world constraints. Using a vector polynomial system and Dixon elimination, the model reduces complexity via decoupling of the position and the altitude. Robotic arm designs with orthogonal or parallel axes are proposed, avoiding the need for intersecting joint axes. Simulations yield eight inverse solutions per target pose within 2ms, with relative errors below 10^{-15} , offering a robust approach for high-precision robotic applications.

Gabriel et al. [14] developed a full kinematic and dynamic

model of a 6-DOF manipulator and designed a computed-torque control (CTC) scheme with a fuzzy-logic extension. The authors derive the robot's D-H parameters to obtain forward/inverse kinematics and use the resulting dynamic equations to implement CTC. A compact fuzzy rule set was applied to improve robustness to disturbances and non-linearities. Simulation in Simulink validates that the fuzzy-CTC achieves precise trajectory tracking under various loads, highlighting that accurate dynamic modelling is essential for high-performance control. Similarly, Ramli et al. [15] modelled a 6-DOF robotic arm based on an ABB IRB series entirely within a simulation environment. They use the RoboAnalyzer software to perform geometric (D-H) forward kinematics and inverse kinematics and then execute dynamic analysis to compute the joint torques required for a chosen motion profile. Their focus is on parametric study and pre-design simulation. When a safety-critical pick-and-place task is given, the simulation yields each joint position and torque over time. This allows selecting suitable motors with desired power before hardware prototyping. They show that the software can generate a simulation of the coordinate for each of the joints, joint parameter, and joint torque to facilitate designing. Badour and Novák [16] bring these concepts to a 5-DOF robot with a vision sensor for 3D-print flaw detection. They find the forward and inverse kinematics from the D-H convention and calculate the Jacobian to examine velocity and force transmission. Singularity analysis is done to determine configurations to be avoided. From the dynamic model, they calculate joint torques and examine energy consumption for planned trajectories. Interestingly, they plan energy-efficient quintic (5th-order) polynomial paths to maneuver the end-effector from print-layer inspection point to point in a smooth manner. Their simulation comparisons between two configurations (elbow-up vs. elbow-down) to attain the same pose are relevant here: the "elbow-down" path has lower peak torques and smoother energy consumption compared to the more aggressive "elbow-up" path. This observation demonstrates how kinematic/dynamic modeling can inform trajectory planning to optimize energy and accuracy. Neeta Sahay et al. [17] presented a detailed study on the simulation of a 2-DOF robot manipulator using the Newton-Euler (N-E) formulation. The dynamic model was derived to relate joint torques with the angular position and velocities of the manipulator's links. Their results showed that a torque ratio of 2.5:1 between the two joints yielded the most balanced and efficient motion profile, suggesting that the methodology could be extended to multi-DOF manipulators for enhanced control accuracy.

Expanding on similar simulation methodologies, Ahsan Kabir Nuhel et al. [18] designed and simulated a 3-DOF manipulator, later extended to a 7-DOF robotic arm for industrial automation using ROS and Gazebo. D-H parameters were used for forward and inverse kinematics, and PID-based control improved trajectory accuracy and reliability, particularly for collaborative robots like the Franka Emika Panda. Carmelo Mineo et al. [19] developed RoboNDT, a MATLAB toolbox for robotic path planning in automated NDT of aerospace

components, addressing curved surface mapping, CAD-real part deviations, and synchronized motion-data acquisition. On the control side, Ali Nihad Wazzan et al. [20] proposed an optimized control scheme for robotic arms by integrating a traditional PID controller with a Fuzzy Logic system and enhancing it using a Social Spider Optimization (SSO) algorithm. Their model, simulated in MATLAB R2021a, aimed to reduce steady-state errors and improve the positional accuracy of robotic arms. Naga et al. [21] found a cost-effective SLAM approach for mobile robots using ultrasonic sensors, an RGB-720p camera, motor drivers, and a Raspberry Pi. This method can enable environment mapping and navigation without prior location knowledge, making it suitable for indoor and outdoor applications such as hospitals and biomedical institutions. Baves Ram S et al. [22] proposed a sensory system which is wearable and can control a 7-DoF robotic arm mimicking human arm motion. Using an IMU, flex sensors, and potentiometers with Arduino and RF communication, the system achieves joint motion replication with less than 2% error, also offers cost-effective applications. Dhanush D et al. [23] presented a robot which is omnidirectional and self-driving indoor elderly surveillance. It features three omni wheels for holonomic motion. It is built on ROS with Jetson Nano, and it enables autonomous navigation and fall/intrusion detection via an onboard camera. NT Muralidharan et al. [24] proposed a flex sensor-based glove system to translate sign language into readable output, enabling communication for individuals with speech or hearing impairments. Using five flex sensors as binary inputs, it supports 32 gestures.

The exploration of this study captures the area of Robotic Kinematics wherein the Forward and Inverse Kinematics operations are integrated with the robotic plant. The objective is to design a system that is capable of decoding the given position vector inputs into the corresponding joint angles by Forward Kinematics block. These joint angles are then driven into the plant and its trajectory is in turn composed into Position vectors of the end effector by the Inverse Kinematics operation which is recorded as output. The input and the output were compared to obtain an estimate of the level of accuracy and precision in the operation of the designed plant.

III. METHODOLOGY

The study outlines the simulation and setup process of a 5 DOF robotic arm with a gripper. The following operation was the calculation of the inverse and forward kinematics of the robotic arm. This was done by simulating the given set of inputs through the kinematics systems to obtain the output of the plant. The parts of the 5 DOF robotic arm were designed in CAD software such that specific shapes could be created with ease, and all these parts were imported into MATLAB software in which the remaining operations were performed. Fig. 3 depicts the CAD model of the Robot Arm employed for the study.

Fig. 1 & Fig. 2 depicts the MATLAB/Simulink model of the entire system, including the robotic arm plant and the kinematics blocks. The plant has a total of 6 revolute joints

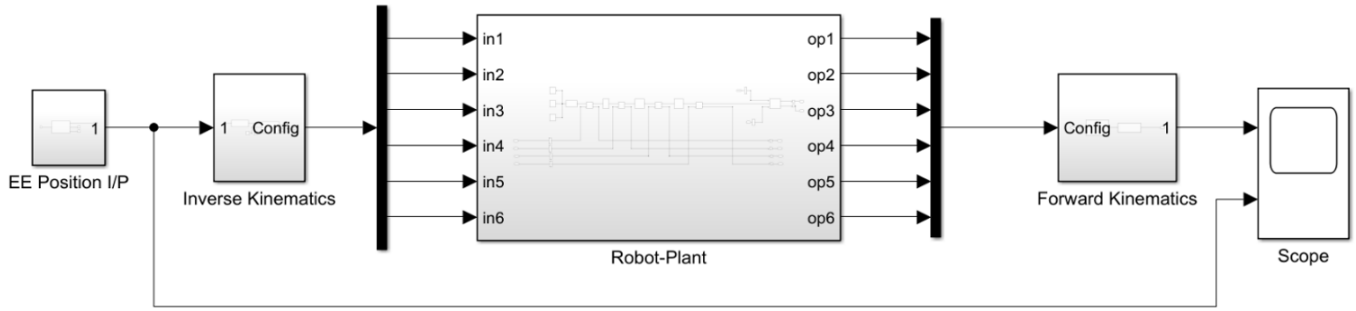


Fig. 1. Simulink Model of the entire system

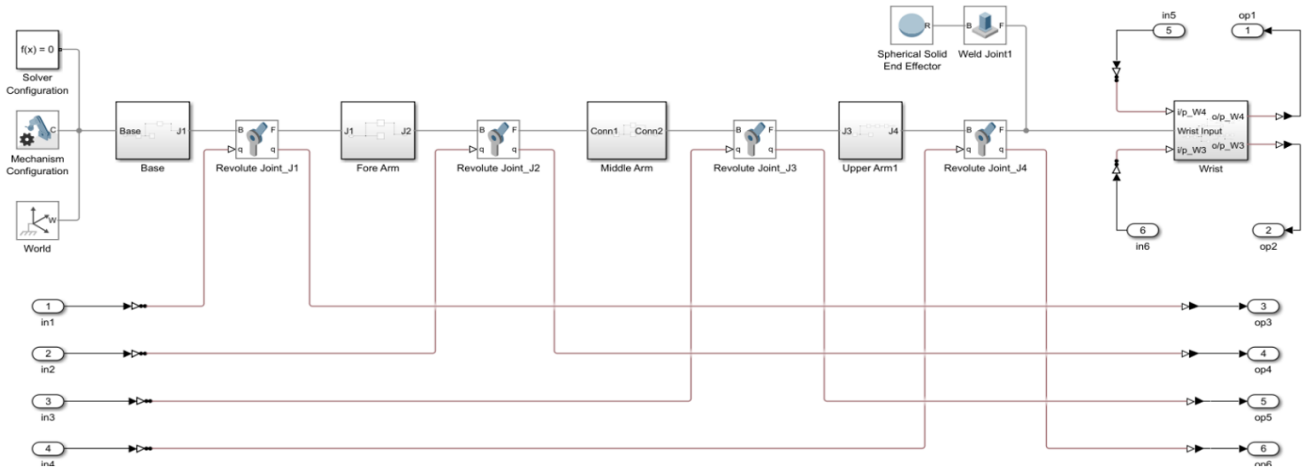


Fig. 2. Simulink Model of the 5-Dof robotic arm

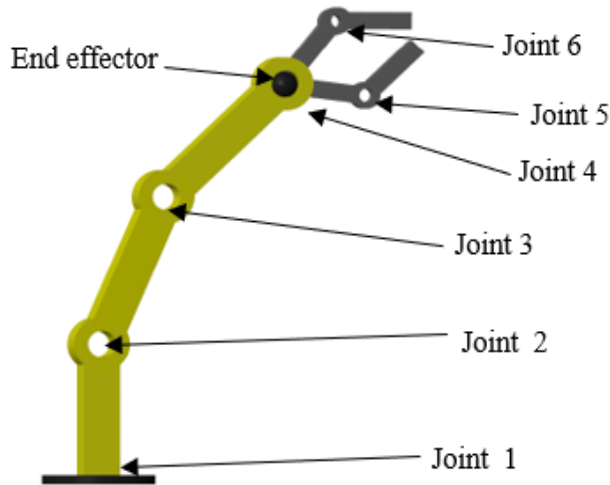


Fig. 3. CAD model of 5-DoF Robot arm

that provide the necessary motion. The intersection of grippers and the third arm is the end effector as indicated in Fig.1. The Waypoints (position vector of the end effector at different time

samples) are entered via the Trapezoidal Velocity Trajectory block as an $n \times 3$ matrix where n rows represent ' n ' number of position vectors $[x, y, z]$ while there are 3 columns each representing coordinates x, y and z respectively. This matrix is then fed into the Inverse Kinematics block that computes the Joint angles corresponding to the given position vectors for all the 6 joints, which then feeds the plant. The output from the plant is driven into the forward kinematics block which gives the various positions of the end effector as the plant traverses the given trajectory. The input and the corresponding output of the end effector are visualized using the scope block. The figures of the graphs are given in the result section.

IV. RESULT

The following graphs in Fig. 4 and Fig. 5 showcase the inputs given to the plant in radians and the respective positions of the end effector obtained through the forward kinetics block. The objective of providing these is to facilitate a qualitative understanding of the forward kinematics for the given plant design. However, the quantitative analysis and observations are made by connecting the velocity profile trajectory block through the inverse kinematics block and then giving the input as position vectors.

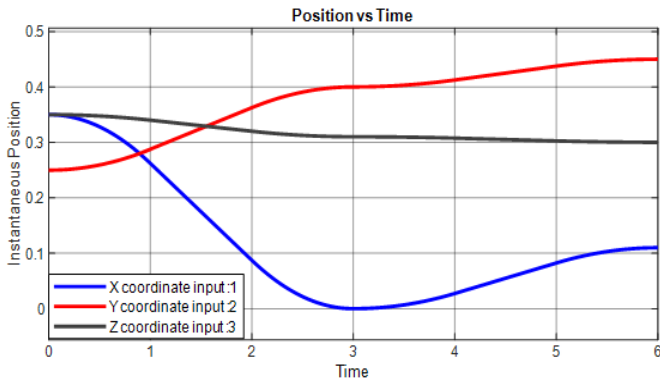


Fig. 4. Coordinates of position vectors as inputs

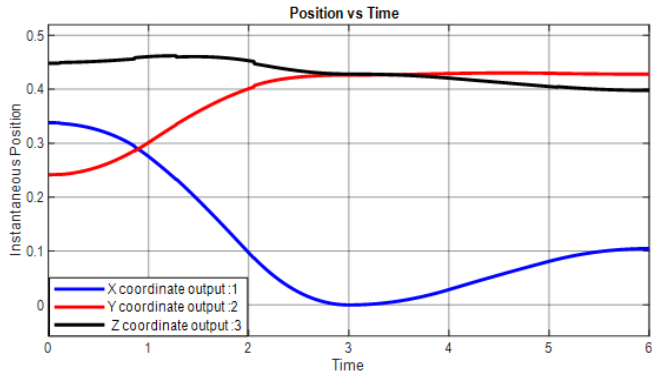


Fig. 5. The trajectory followed by the End Effector relative to the inputs.

The graph of Fig. 4 portrays the desired position provided as input, while the graph in Fig. 5 pictures how well the end effector has followed the given path, where each graph corresponds to the coordinate as specified in the graph.

TABLE 1
INPUT AND OUTPUT COORDINATES FROM THE PLANT

Time	Input positions (in m)			Output positions(in m)		
	X_i	Y_i	Z_i	X_j	Y_j	Z_j
0	0.35	0.25	0.35	0.34	0.24	0.45
1	0.26	0.28	0.34	0.27	0.301	0.45
2	0.08	0.36	0.32	0.09	0.401	0.45
3	0.0	0.4	0.31	0.0	0.43	0.43
4	0.27	0.41	0.31	0.028	0.429	0.42
5	0.08	0.44	0.301	0.08	0.23	0.4

V. DISCUSSION AND ANALYSIS

The observations of the experiment have been tabulated in tables 1,2 and 3.

It can be observed that the output positions of the plant's end effector depicted in the graph of Fig. 4 closely resembles the input position vectors shown in the graph of Fig. 5. However, it is observed from the graph that there is an offset in the end effector's x-, y- and z-coordinates. One of the evident

TABLE 2
DISTANCE OF THE END EFFECTOR FROM ORIGIN AND ERRORS

Time	Input Distance (in m)	Output Distance (in m)	% Error
0	0.555	0.613	10.535
1	0.511	0.612	19.744
2	0.488	0.609	24.815
3	0.506	0.608	20.165
4	0.515	0.601	16.768
5	0.539	0.593	9.949

reasons for this offset can be the inertial effect of the plant. The response due to the plant to the provided input is not an instantaneous process in a practical circumstance and hence is the presence of the offset in the output graph relative to the input. The properties pertaining to the practical situations of the real world like the acceleration due to gravity, mass, and inertia have been imparted to the plant through the mechanical configuration block. Table 3 represents the data where only the Forward Kinematics block is used. Here, the input is directly provided in terms of joint angles and the output is obtained in terms of position vectors of the end effector (the conversion from angles to position vector carried out by the Forward Kinematics block). This table illustrates the working of the Forward Kinematics. Table 1 represents the data obtained from the scenario in which both input and output are in position vector form where the significance of the Inverse Kinematics block is seen, that computes joint angles corresponding to the input position vectors. Table 2 illustrates the precision of the robot arm in its response to the given input. It contains an error percentage column that depicts or quantifies the precision in terms of the error in achieving the desired input positions.

VI. CONCLUSION

This study demonstrates the process of simulation of a 5-DOF robotic arm in MATLAB/Simulink software. It is highly recommendable to create a simulation model and test it through all possible experiments before the practical implementation of the model. The primary focus of this study is to provide manual inputs to the robotic manipulator and analyze the plant's behavior through robotic kinematics to determine the accuracy of its functioning, with respect to the provided input. This is evaluated in terms of the output positions of the end effector.

From the graphs shown in the result section, it can be observed that the robotic manipulator responds perfectly to the given input. Table 2 also indicates that the percentage of error in the robotic manipulator, in following the given input, lies in a tolerable range. This reflects the optimal functioning of the above-designed manipulator.

TABLE 3
END EFFECTOR POSITION WITH RESPECT TO VARIOUS INPUT JOINT ANGLES

Time	Joint Angles						Position of end effector		
	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Xi (m)	Yi (m)	Zi (m)
0	-45	-90	-40	-45	-30	30	14	25	349
1	-20	-60	-20	-35	-50	50	-116	97	355
2	0	0	0	-20	-30	30	-80	149	352
3	20	60	20	-10	0	0	0	98	326
4	45	90	40	0	-30	30	-80	139	190
5	20	60	20	10	-50	50	-170	145	175

VII. FUTURE WORK

This work primarily focuses on the accuracy of the robotic manipulator in achieving the target positions through standalone kinematic models. Future study may extend this work by incorporating additional mechanical parameters such as joint velocities, accelerations, torques etc., to provide more comprehensive performance results. One of the directions for further exploration can be carried out on imparting desired velocity to the plant in order to reach the target position.

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